5.

. In Fig. , is normal to . Two parallel rays enter plane at , , with incidence angle . The refraction angle is . The two refracted rays reach the surface at , , at incidence angles , . In , the distance from to entry is . By the sine rule,

so

Fig.

Likewise, in , the distance from to entry satisfies

so

Varying changes refraction and surface incidence , . At the critical angle given by

total reflection occurs. Substitute into to obtain

Thus

If and , the rays enter, but reach the curved surface at angles above the critical angle , so cannot exit. increases with . From , at , i.e. (normal incidence), is minimal:

At (grazing), . Substitute into to find

. In Fig. , is the - angle; is the - angle. At total reflection,

and

Thus is independent of , i.e. of . At grazing

Fig.